

Submission PHYS1-PS2-SUB06

Question PHYS1-PS2-Q01

notation changes halfway:

1. $v \approx v_i + at \approx 5 + 3(4) \approx 17 \text{ m/s}$

2. $\Delta x \approx v_i t + \frac{1}{2}at^2 \approx 5(4) + \frac{1}{2}(3)(4)^2 \approx 44 \text{ m}$

marks, subscripts, and units are mixed up

result written as: $v=17 \text{ m/s}$; displacement=44 m (please interpret)

Question PHYS1-PS2-Q02

notation changes halfway:

1. $N \approx mg \approx 29.4 \text{ N}$; friction $\approx \mu N \approx 5.9 \text{ N}$

2. $\Sigma F \approx 19 - 5.9 \approx 13.1 \text{ N}$

3. $a \approx \Sigma F/m \approx 4.37 \text{ m/s}^2$

marks, subscripts, and units are mixed up

result written as: $a=4.37 \text{ m/s}^2$; friction opposes motion (please interpret)

Question PHYS1-PS2-Q03

notation changes halfway:

1. $mgh \approx \frac{1}{2}mv^2 \Rightarrow v \approx \sqrt{2gh} \approx \sqrt{2 \cdot 9.8 \cdot 6}$

2. $v \approx 10.84 \text{ m/s}$; mass cancels

marks, subscripts, and units are mixed up

result written as: $v=10.84 \text{ m/s}$ (please interpret)

Question PHYS1-PS2-Q04

notation changes halfway:

1. $p_{\text{total, before}} \approx p_{\text{total, after}}$

2. It applies when external impulse on the system is negligible.

marks, subscripts, and units are mixed up

result written as: total momentum before equals after when external impulse is negligible (please interpret)

Question PHYS1-PS2-Q05

notation changes halfway:

1. Impulse is area under $F(t)$: $J \approx F\Delta t \approx 6(0.5) \approx 3 \text{ N}\cdot\text{s}$.

2. $\Delta p \approx J$, so momentum changes by $3 \text{ kg}\cdot\text{m/s}$ in the force direction.

marks, subscripts, and units are mixed up

result written as: impulse=3 N·s; momentum changes by $3 \text{ kg}\cdot\text{m/s}$ (please interpret)

Question PHYS1-PS2-Q06

notation changes halfway:

1. $v \approx v_i + at \approx 4 + 2(5) \approx 14 \text{ m/s}$

2. $\Delta x \approx v_i t + \frac{1}{2}at^2 \approx 4(5) + \frac{1}{2}(2)(5)^2 \approx 45 \text{ m}$

marks, subscripts, and units are mixed up

result written as: $v=14 \text{ m/s}$; displacement=45 m (please interpret)

Question PHYS1-PS2-Q07

notation changes halfway:

1. $N \approx mg \approx 19.6 \text{ N}$; friction $\approx \mu N \approx 3.9 \text{ N}$

2. $\Sigma F \approx 11 - 3.9 \approx 7.1 \text{ N}$

3. $a \approx \Sigma F/m \approx 3.54 \text{ m/s}^2$

marks, subscripts, and units are mixed up

result written as: $a=3.54 \text{ m/s}^2$; friction opposes motion (please interpret)

Question PHYS1-PS2-Q08

notation changes halfway:

1. $mgh \approx \frac{1}{2}mv^2 \Rightarrow v \approx \sqrt{2gh} \approx \sqrt{2 \cdot 9.8 \cdot 6}$

2. $v \approx 10.84 \text{ m/s}$; mass cancels

marks, subscripts, and units are mixed up

result written as: $v=10.84 \text{ m/s}$ (please interpret)